Curriculum Vitae of Rebecca Gatt

Computational Physicist | Scientific Modelling | Numerical Simulation | Statistical Inference | Machine Learning

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Current Affiliation: Institute of Space Sciences and Astronomy, University of Malta, Msida

Current Position: Post-Doc at the Institute of Space Sciences and Astronomy

Professional Profile

Computational physicist with a PhD in cosmology and research experience spanning numerical modelling, scientific software development, statistical inference and machine learning. Current work focuses on data-driven reconstruction of physical models using genetic algorithms, Monte Carlo Markov Chains, Bayesian inference and machine learning, including the development and modification of scientific simulation codes. Experienced in independent research, collaborative international projects, scientific programming and communicating technical results to specialist and non-specialist audiences. Interested in applying these computational and modelling skills to space environments, radiation transport, simulation and spacecraft effects analysis.

Research Experience

- *Nov 2024 – Present*: **Post-Doctoral Research Support Officer**
  Institute of Space Sciences and Astronomy, University of Malta, Msida
  - Project: *Cosmic Leaning Project: Building Learned Cosmological Models in the Age of Big Survey Data*
  - *Advisor*: Prof. Levi Said
  - *Focus:* Data-driven reconstruction of cosmological models integrating Genetic Algorithms, Bayesian inference, and machine learning. Modifying CLASS for modified gravity frameworks.

- *Mar 2026 – Present*: **Researcher Part-time**
  Institute of Applied Science, MCAST, Paola
  - Project: *CosmoAI*
  - *Advisor*: Mr. Mark Pace
  - *Focus:* The project aims to build robust data-driven pipelines for extracting physical models from survey data, resolving current tensions in observational cosmology, and contributing to computational efficiency within broader scientific domains.

Education

- *Oct 2020 – Sep 2025:* **Ph.D., Institute of Space Sciences and Astronomy, University of Malta**
  - Thesis: *Cosmological Constraints in Physics beyond General Relativity*
  - *Supervisor*: Prof. Levi Said
  - *Submission*: 30th Sep 2024 | Defense: 12th Dec 2024
  - *Focus*: Modified gravity theories (Teleparallel Gravity), computational and statistical methods for cosmology.

- **B.Sc. (Hons) in Mathematics and Physics, University of Malta**
  Oct 2016 – Jun 2020
  - *Thesis*: *Testing Gravity Theories using Cosmological Observations*
  - *Supervisor*: Prof. Levi Said

Areas of Specialization

Physics | Theoretical & Computational Cosmology | Dark Energy | Modified Gravity | Cosmological Tensions Gravity | Machine Learning

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Publications

Peer-Reviewed Journals:

1. **R. Briffa**, C. Escamilla-Rivera, J. Levi Said, and J. Mifsud (2024), Growth of structures using redshift space distortion in f(T) cosmology, *Mon. Not. Roy. Astron. Soc.*, 528, 2711–2727.

2. R. Sandoval-Orozco, C Escamilla-Rivera, **R. Briffa**, and J. Levi Said (2024), Testing f(T) cosmologies with HII Hubble diagram and CMB distance priors, *Phys. Dark Univ.,* 16, 101641.

3. D. Aguiar Gomes, **R. Briffa**, A. Kozak, J. Levi Said, M. Saal, and A. Wojnar, Cosmological constraints of Palatini f(R) gravity (2024), *J. Cosmol. Astropart. Phys.*, 01, 011.

4. R. Sandoval-Orozco, C. Escamilla-Rivera, **R. Briffa**, and J. Levi Said. (2024), f(T) cosmology in the regime of quasar observations, *Phys. Dark Univ,.* 43, 101407

5. **R. Briffa,** C. Escamilla-Rivera, J. Levi Said, and J. Mifsud (2023), f(T,B) Gravity in the late Universe in the context of local measurements, *Phys. Dark Univ,.* 39, 101153.

6. **R. Briffa**, S. Capozziello, J.L. Said, J. Mifsud and E.N. Saridakis (2021), Constraining teleparallel gravity through Gaussian processes, *Class. Quantum Grav*, 38, 055007.

7. **R. Briffa**, C. Escamilla-Rivera, J. Levi Said, J. Mifsud and N.L. Pullicino (2022), Impact of H0 priors on f(T) late time cosmology, *Eur. Phys. J.Plus,* 137, 532.

8. **R. Briffa**, C. Escamilla-Rivera, J. Levi Said, and J. Mifsud (2024), Constraints on f(T) cosmology with Pantheon+, *Mon. Not. Roy. Astron. Soc.*, 552, 6024–6034.
9. E Di Valentino et al. (2025), The CosmoVerse White Paper: Addressing observational tensions in cosmology with systematics and fundamental physics, *Phys. Dark Univ.,* 49, 101965.
10. *S. Bahamonde, **R. Briffa,** K. Dialektopoulos, D. Iosifidis, J. Levi Said (2025), Cosmology of Cubic Poincare Gauge gravity, J. Cosmol. Astropart. Phys.*.

Manuscripts in Preparation:

- **R. Briffa**, J. Levi Said, (2025), Confronting Primordial Gravitational Waves with f(T) gravity (in prep.)
- **R. Briffa**, E. Jensko , and L. Csillag, Cosmological constraints on purely kinetic k-essence, (2025), (in prep.)
- **R. Briffa**, E. Jensko , O. Akarsu et al, Sign-changing dark energy from modified gravity (2025), (in prep.)

Conferences & Talks

- Cosmoverse@Bilbao, Bilbao, Spain – Sept 2026, An Agnostic Exploration of Novel Equation Structures for Dynamical Dark Energy
- IBS Workshop: Gravity and Cosmology by the Sea, Corfu, Greece, Sept 2026 – **Invited Speaker**: An Agnostic Exploration of Novel Equation Structures
- COSMO'26, Leiden University, Netherlands, August 2026
- Max-Planck-Institut für Gravitationsphysik, Albert-Einstein-Institut, March 2026– Invited Speaker at the Journal Club: Can we learn Cosmological Models from Data?
- Cosmology Seminars @Szczecin, University of Szczecin, May 2026 – Invited Speaker at the Journal Club: Can we learn Cosmological Models from Data?
- Tensions in Cosmology, Corfu, Greece, Sep 2025 – Addressing Cosmic Tensions with f(T) Gravity
- A decade of ISSA, University of Malta, Sep 2024 – Constraining parameters in Teleparallel Gravity
- COSMO'24, Kyoto University, Japan, Oct 2024 – Constraining parameters in Teleparallel Gravity
- CosmoVerse@Krakow, Jagiellonian University, Poland, Jul 2024 – Constraining parameters in f(T) gravity

- Standard cosmology at the threshold of change?, University of Thessaloniki, Greece, Jun 2024 – Addressing Cosmic Tensions with f(T) Gravity
- Iberian Cosmology Meeting (IberiCOS 2024), Salamanca, Spain, Mar 2024 – Late-time Accelerating Universe and Growth of Structure in Teleparallel Gravity
- CosmoVerse@Lisbon, Lisbon, Portugal, May–Jun 2023 – Constraints in Cosmological parameters in Teleparallel Gravity
- Tensions in Cosmology, Corfu, Greece, Sep 2022 – Late-time Acceleration in Teleparallel Gravity
- ALTECOSMOFUN'21, Online, Sep 2021 – Impact of H0 priors on f(T) late-time cosmology
- Geometric Foundations of Gravity, Online, Jun–Jul 2021 – Constraints in f(T) gravity
- Iberian Cosmology Meeting (IberiCOS 21), Online, Mar 2021 – Constraining Teleparallel Gravity through Gaussian Processes

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Academic Visits

- **Universidad Nacional Autónoma de México**, 2023 – Research visit at Instituto de Ciencias Nucleares. Presented seminar, collaborated on MCMC and HiClass datasets.

- **Università degli Studi di Torino**, 2023 – Short-term scientific mission under Dr. Francesco Pace, applying Horndeski theory to Teleparallel gravity.
- **Max-Planck-Institut für Gravitationsphysik**, 2026 – Research visit at the Albert Einstein Institute, working with Dr. Reginal Bernardo on Genetic Algorithms applied to cosmological models.

Computational Skills

Python | Mathematica | CLASS | HiClass | MontePython | Cobaya | getdist | C++

Professional Activities

- Peer Reviewer: Journal of High Energy Astrophysics, European Physical Journal Plus, Scientific Reports,

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Teaching Experience

- PHY3111: Mathematics for Physicists 1, University of Malta (2022–2025)
- PHY2235: Experimental Physics, University of Malta (2020–2024)

- PHY1119: Fundamentals of Experimental Physics, University of Malta (2020–2024)
- PHY2230: Experimental Physics, University of Malta (2020–2024)
- PHY1105: Fundamentals of Experimental Physics, University of Malta (2020–2024)
- DGP0982: Introduction to LaTeX for Professional and Scientific Writing, University of Malta (2021–2024)

Memberships

- Member, CosmoVerse Action (CA21136)
- Management Committee, FuSE Action (CA24101)

Awards

- Dean's List: 2017/18, 2018/19

Mentoring

- Supervised 1 undergraduate and 2 Master's students on research projects.

Public Engagement & Outreach

- Unconventional Careers: Promoting diverse careers in science for schools
- Science in the City: Public science festival participation
- School Visits: Inspiring young students and discussing scientific concepts
- ASM events: Networking and sharing research at Esplora Science Centre

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